

Linear Systems and Elimination

Historical Note

The oldest surviving account of the method in this chapter is Chapter 8 of the Chinese classic *Jiuzhang Suanshu*, the *Nine Chapters on the Mathematical Art*, compiled by roughly the first century of the common era. That chapter is called *fangcheng*, meaning “rectangular arrays”. Its author lays out the coefficients of a system as a square of counting rods on a board and then subtracts multiples of one column from another until a single unknown is left standing. That is exactly the algorithm below, written two thousand years before matrices had a name. In Europe the procedure appears in Newton’s lecture notes of about 1670, printed in 1707 as the *Arithmetica Universalis*, and Gauss used it in 1810 while reducing the observations of the asteroid Pallas by least squares. The label “Gaussian elimination” is a twentieth-century invention of numerical analysts and, as Grcar [272] documents at length, it credits the wrong person by about two millennia. We keep the name because everyone does, but it is worth knowing whose method it is.

Chapter 2 ended with a question we could only answer by guessing: given vectors $v_1, \dots, v_k$ and a target b , is b a linear combination of the v_i ? In Example 2.6 we found the coefficients by rearranging two equations by hand. That works for two equations. It does not work for two hundred. This chapter turns the guessing into an algorithm, and in doing so produces the single most-executed procedure in all of applied mathematics.

3.1 Systems, and what a solution is

Definition 3.1 (Linear system). A system of m linear equations in n unknowns

$x_1, \dots, x_n$ is a list

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1,$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2,$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m,$$

with the a_{ij} and b_i given real numbers. A solution is a vector $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ satisfying every equation at once. The solution set is the set of all of them. The system is consistent if that set is nonempty and inconsistent otherwise. It is homogeneous if every b_i is zero.

The word *linear* is doing real work here. Each unknown appears to the first power only, multiplied by a constant, and never multiplied by another unknown. No x_1^2 , no x_1x_2 , no $\sin x_1$. That restriction is what makes a complete theory possible, as Chapter 1 argued.

Key Idea

There are three ways to read the same system, and fluency means switching between them without thinking.

Row picture. Each equation describes a flat set: a line in $\mathbb{R}^2$, a plane in $\mathbb{R}^3$, a hyperplane in general. A solution is a point lying on all of them at once. Solving is intersecting.

Column picture. Writing $A_1, \dots, A_n$ for the columns of coefficients, the system says $x_1A_1 + \cdots + x_nA_n = b$. A solution is a recipe for building b out of the columns. Solving is asking whether $b \in \text{span}\{A_1, \dots, A_n\}$.

Map picture. The system is $Ax = b$ for the linear map A of Chapter 1. Solving is asking what A sends to b .

The row picture is where geometry lives, the column picture is where linear algebra lives, and the map picture is where the rest of this book lives.

Two equations, three pictures

Consider

$$x + y = 3, \quad x - y = 1.$$

The row picture draws two lines in the plane; they cross at $(2, 1)$. The column picture asks for $x \begin{pmatrix} 1 \\ 1 \end{pmatrix} + y \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, and indeed $2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$. Same answer, different question.

Two lines in the plane can do exactly three things, and this already exhausts the possibilities for linear systems of any size.

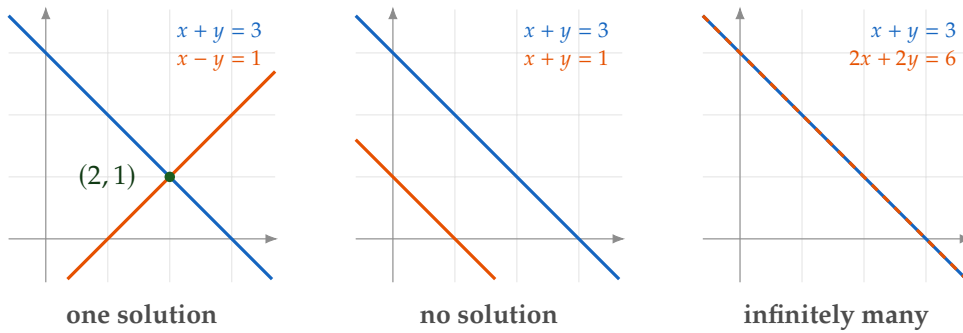


Figure 3.1. The only three things two lines in the plane can do. They meet once, they never meet, or they are the same line. Every linear system, of any size over $\mathbb{R}$, has exactly one of these three outcomes: a unique solution, none, or infinitely many. In the third panel the two equations describe the same line, so the dashed orange line is drawn on top of the solid blue one deliberately.

Theorem 3.2 (The trichotomy). *A system of linear equations over $\mathbb{R}$ has either no solution, exactly one solution, or infinitely many. In particular it can never have exactly two.*

Proof. Suppose the system $Ax = b$ has two distinct solutions $u \neq v$. For any $t \in \mathbb{R}$ put $w_t = u + t(v - u)$. By linearity of each equation,

$$Aw_t = Au + t(Av - Au) = b + t(b - b) = b,$$

so every w_t is a solution. The map $t \mapsto w_t$ is injective because $v - u \neq 0$, so we have produced infinitely many distinct solutions. Hence two solutions force infinitely many, which leaves only the three stated cases. $\square$

That short argument is worth pausing over, because it uses nothing but linearity and yet settles a question that would be hopeless for polynomial equations. It is a first taste of what Chapter 1 promised: linear problems are rigid, and rigidity is what lets us classify.

Figure 3.2 repeats the three outcomes one dimension up, where the equations are planes rather than lines, and separates the two questions the picture answers: the coefficient matrix decides how many solutions there are if there are any, and the right-hand side decides whether there are any at all.

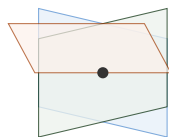
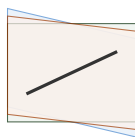
one point: $\det \neq 0$ a line: $\det = 0$, consistentnothing: $\det = 0$, inconsistent

Figure 3.2. The trichotomy of Chapter 3, one dimension up. Two lines in the plane meet in a point, a whole line, or nowhere; three planes in space do the same, and the algebra is identical. A unique solution needs the three normal vectors to be independent, so $\det \neq 0$. When they are dependent the determinant vanishes and two cases remain, separated not by the matrix but by the right-hand side: if the system is consistent the solutions form a line, or a plane if the dependence is worse, and if it is inconsistent there are none. *The coefficient matrix decides how many solutions there are if there are any; the right-hand side decides whether there are any at all, and that split is the content of the particular-plus-homogeneous description.*

3.2 Row operations, and why they are safe

We solve a system by replacing it with a simpler system having the same solutions. Three moves are permitted.

Definition 3.3 (Elementary row operations). *Given a system, the following operations are called elementary:*

(R1) **Swap**: exchange two equations.

(R2) **Scale**: multiply an equation by a nonzero constant λ .

(R3) **Replace**: add a multiple of one equation to a different equation.

Proposition 3.4 (Elementary operations preserve the solution set). *If a system S' is obtained from a system S by one elementary row operation, then S and S' have exactly the same solution set.*

Proof. For (R1) nothing changes but the order in which we list conditions, and a conjunction does not care about order.

For (R2), suppose equation i reads $L_i(x) = b_i$ and is replaced by $\lambda L_i(x) = \lambda b_i$ with $\lambda \neq 0$. If $L_i(x) = b_i$ then multiplying both sides gives the new equation. Conversely, if $\lambda L_i(x) = \lambda b_i$ then dividing by λ , which is legitimate precisely because $\lambda \neq 0$, recovers the old one. All other equations are untouched.

For (R3), suppose equation j is replaced by $L_j(x) + \mu L_i(x) = b_j + \mu b_i$, with $i \neq j$.

If x solves S then it satisfies both $L_i(x) = b_i$ and $L_j(x) = b_j$, hence the combination. Conversely, if x solves S' then it satisfies the new equation j and the untouched equation i ; subtracting μ times the latter from the former returns $L_j(x) = b_j$. The operation is undone by adding $-\mu$ times equation i , which is why the argument is symmetric. $\square$

Caution

The conditions in (R2) and (R3) are not fussiness. Scaling by $\lambda = 0$ destroys an equation and can only enlarge the solution set: from $x = 1$ it manufactures $0 = 0$, which everything satisfies. Adding a multiple of an equation *to itself* is the same trap in disguise, since adding -1 times equation i to equation i annihilates it. Each permitted operation is *reversible*, and reversibility is exactly why no solutions are gained or lost.

Key Idea

Every elementary operation is reversible, and its inverse is an operation of the same kind. Swapping twice restores the order; scaling by λ is undone by scaling by $1/\lambda$; adding μ times row i is undone by adding $-\mu$ times row i . This is the whole reason elimination is trustworthy, and in Chapter 4 it will reappear as the statement that elementary matrices are invertible.

3.3 The augmented matrix

Writing x, y, z and plus signs a hundred times is wasted effort: the symbols never change, only the numbers do. So we drop the symbols.

Definition 3.5 (Coefficient and augmented matrix). *For the system of Definition 3.1, the coefficient matrix is the $m \times n$ array $A = (a_{ij})$, and the augmented matrix is A with the right-hand side attached as an extra column, written $[A \mid b]$.*

The system

$$\begin{array}{rcl} x + y + z = 6, \\ 2x - y + 3z = 9, \\ -x + 4y - 2z = 1, \end{array} \quad \text{becomes} \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 3 & 9 \\ -1 & 4 & -2 & 1 \end{array} \right].$$

The vertical rule is a reminder, not an operation: the last column is data, not a coefficient. Row operations act on whole rows of this array, augmented column included.

3.4 Echelon form and the elimination algorithm

Definition 3.6 (Row echelon form). A matrix is in row echelon form if

- (i) every row consisting entirely of zeros lies below every nonzero row, and
- (ii) in each nonzero row, the leftmost nonzero entry, called the pivot, lies strictly to the right of the pivot of the row above it.

A column containing a pivot is a pivot column; the others are free columns. The matrix is in reduced row echelon form, or RREF, if in addition every pivot equals 1 and is the only nonzero entry in its column.

Condition (ii) is what produces the characteristic staircase: each row starts later than the last, and the steps may be wider than one column but never narrower. Figure 3.3 shows the shape, with the pivots marked and the whole region below and to the left of them forced to zero.

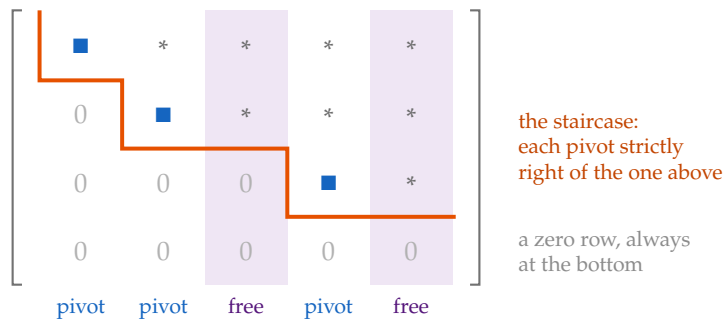


Figure 3.3. Row echelon form. Filled squares are pivots, asterisks are entries that may be anything, and everything below and to the left of the orange staircase is zero. Columns 3 and 5 carry no pivot, so the corresponding unknowns will be free. The steps may be wide, as between the second and third pivots, but they always move to the right.

Key Idea

Gaussian elimination. Work left to right. In the current column, find a nonzero entry at or below the current row; if there is none, move one column right and try again. Otherwise swap it into position, call it the pivot, and use (R3) to subtract multiples of the pivot row from every row beneath so that the whole column below the pivot becomes zero. Advance one row and one column and repeat. The process stops after at most $\min(m, n)$ pivots, and the result is a matrix in row

echelon form.

Example 3.7 (Elimination, in full). We solve the system above. Start from the augmented matrix and clear the first column, using $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 + R_1$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 3 & 9 \\ -1 & 4 & -2 & 1 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & 1 & -3 \\ 0 & 5 & -1 & 7 \end{array} \right].$$

The pivot of the second row is -3 . Clear below it with $R_3 \rightarrow R_3 + \frac{5}{3}R_2$:

$$\longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & 1 & -3 \\ 0 & 0 & \frac{2}{3} & 2 \end{array} \right].$$

This is row echelon form, with three pivots $1, -3, \frac{2}{3}$ in columns 1, 2, 3. Now read the equations back and work upwards, which is called back substitution:

$$\frac{2}{3}z = 2 \implies z = 3, \quad -3y + z = -3 \implies y = 2,$$

$$x + y + z = 6 \implies x = 6 - 2 - 3 = 1.$$

The solution is $(x, y, z) = (1, 2, 3)$. Substituting back into the original three equations gives $1 + 2 + 3 = 6$, $2 - 2 + 9 = 9$ and $-1 + 8 - 6 = 1$. Always perform this check: it costs a few seconds and catches nearly every arithmetic slip.

Example 3.8 (The same system, driven to RREF). Instead of back substituting, we may keep going with row operations. Scale the pivots to 1 by $R_2 \rightarrow -\frac{1}{3}R_2$ and $R_3 \rightarrow \frac{3}{2}R_3$, then clear above each pivot:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -\frac{1}{3} & 1 \\ 0 & 0 & 1 & 3 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right].$$

In the first arrow we used $R_1 \rightarrow R_1 - R_3$ and $R_2 \rightarrow R_2 + \frac{1}{3}R_3$; in the second, $R_1 \rightarrow R_1 - R_2$. The final array reads $x = 1, y = 2, z = 3$ with no work left to do. The extra effort spent reaching RREF is exactly the effort saved on back substitution, so the two routes cost about the same; RREF wins when we want the answer in a form a machine can read off, and back substitution wins when we are doing it by hand.

Theorem 3.9 (Uniqueness of the reduced form). *Every matrix can be brought to reduced row echelon form by elementary row operations, and the reduced form is unique: it does not depend on which sequence of operations was used. In particular the set of pivot columns, and hence the number of pivots, is an invariant of the original matrix.*

Existence is the algorithm above followed by upward clearing. Uniqueness is a genuine theorem and we defer its proof to Chapter 4, where row operations become matrix multiplication and the statement becomes transparent. Its consequence, though, we use immediately: the number of pivots is a well-defined number attached to a matrix. That number is the *rank*, and it will run through the rest of the book.

3.5 Reading off the answer

Once the augmented matrix $[A \mid b]$ is in echelon form, the entire solution set can be read from the pivot pattern.

Theorem 3.10 (Consistency and the shape of the solution set). *Let $[A \mid b]$ be brought to row echelon form, and let r be the number of pivots of A .*

- (a) *The system is inconsistent exactly when the augmented column contains a pivot, that is, when some row reads $[0 \cdots 0 \mid c]$ with $c \neq 0$.*
- (b) *If it is consistent and $r = n$, the solution is unique.*
- (c) *If it is consistent and $r < n$, the solution set is described by $n - r$ free parameters, one for each free column, and is therefore infinite.*

Proof. (a) A row $[0 \cdots 0 \mid c]$ is the equation $0 = c$, which no x satisfies, and by Proposition 3.4 the reduced system has the same solutions as the original. Conversely, if no such row occurs, assign arbitrary values to the free unknowns and solve upward: each pivot row determines its pivot unknown from those to its right, so a solution exists.

(b) and (c) follow from that construction. There are $n - r$ free unknowns, and the back substitution determines the remaining r uniquely once the free ones are fixed. So the solutions are in bijection with $\mathbb{R}^{n-r}$: a single point when $r = n$, and a set of infinite size otherwise. $\square$

Example 3.11 (An inconsistent system). *Take $x + y = 2$ and $2x + 2y = 5$. One step, $R_2 \rightarrow R_2 - 2R_1$, gives*

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 2 & 2 & 5 \end{array} \right] \longrightarrow \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & 1 \end{array} \right],$$

whose second row asserts $0 = 1$. There is no solution. The row picture is the middle panel of Figure 3.1: two parallel lines. The column picture is just as telling, since both columns of A are multiples of $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, so their span is a single line, and $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$ is not on it.

Example 3.12 (A system with a free variable). Take

$$x + 2y - z = 3, \quad 2x + 4y + z = 9.$$

Eliminating with $R_2 \rightarrow R_2 - 2R_1$:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 9 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & 3 \end{array} \right].$$

Pivots sit in columns 1 and 3, so $r = 2$ while $n = 3$: column 2 carries no pivot, and Theorem 3.10 promises $n - r = 1$ free parameter. The second row reads $3z = 3$, so $z = 1$. Substituting into the first row, $x + 2y - 1 = 3$, hence $x = 4 - 2y$. Only y is unconstrained, so write $y = t$ and collect:

$$(x, y, z) = (4 - 2t, t, 1) = (4, 0, 1) + t(-2, 1, 0), \quad t \in \mathbb{R}.$$

The solution set is a line in $\mathbb{R}^3$: it passes through $(4, 0, 1)$ and runs in the direction $(-2, 1, 0)$. Substituting back confirms it for every t at once, which is the only check worth doing: $(4 - 2t) + 2t - 1 = 3$ and $2(4 - 2t) + 4t + 1 = 9$, both identically in t . Notice how the two parts of the answer arrive from different places. The point $(4, 0, 1)$ comes from the right-hand side; the direction $(-2, 1, 0)$ comes from the free column and would be there even if the right-hand side were zero. Section 3.6 explains why that split is universal.

Caution

Which unknowns are free is decided by the pivot pattern, not by taste. In Example 3.12 it is y that is free, because column 2 carries no pivot; declaring z free instead and solving for y would have been an error, since the second equation pins z down completely. Read the free columns off the echelon form before you name any parameter.

3.6 Homogeneous systems, and the structure of every solution set

Set every right-hand side to zero. A homogeneous system $Ax = 0$ is never inconsistent, since $x = 0$ always works; the question is whether anything else does.

Proposition 3.13 (Wide homogeneous systems always have a nonzero solution). *If A is $m \times n$ with $m < n$, then $Ax = 0$ has a solution other than $x = 0$.*

Proof. Elimination produces at most one pivot per row, so $r \leq m < n$. By Theorem 3.10(c) there are $n - r \geq 1$ free parameters, and setting one of them to 1 yields a solution that is not the zero vector. $\square$

Read in the column picture, this says that more vectors than dimensions must be linearly dependent, which is the fact underlying the whole theory of dimension in Chapter 6. Read practically, it says an underdetermined problem always has a nontrivial null direction, which is why fitting a model with more parameters than data points can never pin the parameters down.

Theorem 3.14 (Particular plus homogeneous). *Suppose $Ax = b$ is consistent and p is any one solution. Then the full solution set is*

$$\{p + h : Ah = 0\}.$$

That is, the solution set of $Ax = b$ is the solution set of $Ax = 0$ translated by p .

Proof. If $Ah = 0$ then $A(p + h) = Ap + Ah = b + 0 = b$, so every such vector solves the system. Conversely, if $Ax = b$ then $A(x - p) = b - b = 0$, so $h = x - p$ solves the homogeneous system and $x = p + h$ has the stated form. $\square$

Key Idea

Every solution set of a linear system is a flat object: a point, a line, a plane, or a higher-dimensional analogue, sitting somewhere in $\mathbb{R}^n$ and not necessarily through the origin. It passes through the origin exactly when the system is homogeneous. The homogeneous solutions supply the *shape*, and any single particular solution supplies the *position*.

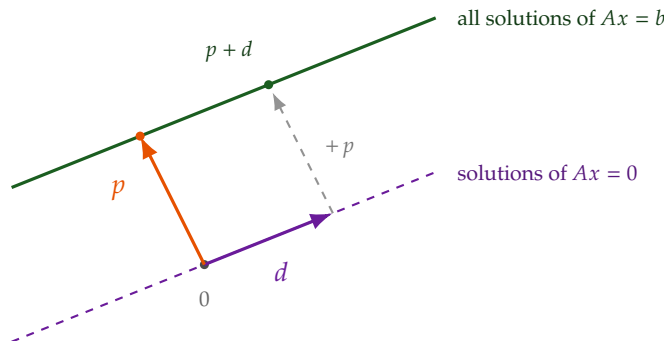


Figure 3.4. Theorem 3.14 in one picture. The dashed line is the solution set of the homogeneous system, which always passes through the origin. The solid line is the solution set of $Ax = b$: the same line, shifted by any one particular solution p . Translating the point d by p lands on $p + d$, and every point of one line arises from a point of the other in exactly this way.

Example 3.15 (Structure, verified). Return to Example 3.12. Its homogeneous version is $x + 2y - z = 0$, $2x + 4y + z = 0$; the same elimination gives $3z = 0$, so $z = 0$, and then $x = -2y$. The homogeneous solutions are therefore $t(-2, 1, 0)$, a line through the origin. The full solution set we found was $(4, 0, 1) + t(-2, 1, 0)$: the same line, shifted by the particular solution $p = (4, 0, 1)$, exactly as Theorem 3.14 predicts. Note that p is not canonical. Taking $t = 1$ instead gives $p' = (2, 1, 1)$, and $(2, 1, 1) + t(-2, 1, 0)$ describes the identical set. Position can be reported from any point on the line.

3.7 What elimination costs

Elimination is not merely a proof device; it is what a computer actually runs, so its price matters.

Proposition 3.16 (Operation count). Forward elimination on a dense $n \times n$ system requires about $\frac{1}{3}n^3$ multiplications and about as many additions, so roughly $\frac{2}{3}n^3$ arithmetic operations in total. Back substitution costs a further $\frac{1}{2}n^2$ multiplications, which is negligible by comparison.

Sketch. At step k there are $n - k$ rows below the pivot, and updating each of them touches $n - k + 1$ entries, one of them the augmented column. That is about $(n - k)^2$ multiplications, and summing over $k = 1, \dots, n$ gives $\sum_{j=1}^{n-1} j^2 \approx n^3/3$. Each multiplication is paired with a subtraction, which doubles the count. $\square$

Insight

The cubic term is the whole story of numerical linear algebra. Doubling n multiplies the work by eight. A dense system with $n = 1000$ needs about 7×10^8 operations and finishes in well under a second on a laptop. A dense system with $n = 10^6$, the size of a routine problem in scientific computing or a moderate machine-learning model, would need about 7×10^{17} operations and is simply out of reach. This is why the subject splits in two. When the matrix is small or dense, we eliminate. When it is large, we exploit structure: sparsity, symmetry, low rank, or a fast matrix-vector product, and we solve iteratively, approaching the answer rather than computing it exactly. Everything in Parts III and IV about positivity, spectra and conditioning exists to make that second path work.

Caution

Mathematically, any nonzero entry may serve as a pivot. Numerically, this is false. Consider

$$0.0001x + y = 1, \quad x + y = 2,$$

whose exact solution is $x = 1/0.9999 \approx 1.0001$, $y \approx 0.9999$. Eliminating with the tiny pivot 0.0001 multiplies row one by 10,000 and subtracts, producing $-9999y = -9998$. In arithmetic carrying only three significant digits both of those round to -1.00×10^4 , giving $y = 1.00$; back substitution then yields $x = (1 - 1.00)/0.0001 = 0$, which is not close to 1. Swapping the rows first, so that the pivot is the larger entry 1, gives $y = 1.00$ and $x = 1.00$, both correct to three digits. The rule, called *partial pivoting*, is to choose as pivot the largest available entry in the column. It costs almost nothing and it is why library solvers are reliable; see Trefethen and Bau [607].

3.8 What we can now answer

The question Chapter 2 left open is settled. To decide whether $b \in \text{span}\{v_1, \dots, v_k\}$, place the v_i as the columns of a matrix A , eliminate on $[A \mid b]$, and look: a pivot in the augmented column means no, and its absence means yes, with the coefficients read off by back substitution. If there are free columns the recipe is not unique, and Theorem 3.14 describes all of them at once.

Three further questions are now within reach, and they organise the next three chapters. What does A have to look like for $Ax = b$ to be solvable for *every* b ? When can the elimination be recorded once and replayed cheaply for many different b ? And what is the meaning of the number of pivots, which Theorem 3.9 says is intrinsic? The

answers are invertibility, the LU factorisation, and rank.

Open Problem 3.17 (Solving a sparse system in nearly linear time). *Elimination on an $n \times n$ system costs $O(n^3)$, and even when the matrix has only m nonzero entries the fill-in generated along the way can destroy that sparsity entirely. Is there an algorithm that solves $Ax = b$ for an arbitrary sparse nonsingular A to accuracy ε in $O(m \cdot \text{polylog}(n/\varepsilon))$ time?*

What is known. For one important class, the answer is yes. Spielman and Teng showed in 2004 that systems whose matrix is symmetric and diagonally dominant, which includes every graph Laplacian, can be solved in nearly linear time [578], and the constants have since become practical. Nothing comparable is known for general sparse matrices, symmetric or otherwise. The obstruction is not merely technical: the Laplacian solvers work by sparsifying, replacing the graph by a sparser one with nearly the same quadratic form, and it is the graph structure that makes sparsification possible.

Where the difficulty sits. Elimination is a sequence of rank-one updates, and each one can turn a sparse row into a dense one. Choosing an elimination order that avoids this is exactly the minimum fill-in problem of Chapter 113, which is NP-hard. So a nearly linear algorithm cannot be an ordering heuristic; it has to give up on exact elimination and iterate towards the answer, which means it needs a preconditioner, which means it needs a notion of approximating a matrix by a sparser one that survives inversion.

Where a reader might start. Take the $n \times n$ tridiagonal matrix of the second difference and eliminate it in order: no fill-in at all. Now permute the rows and columns at random and eliminate again, and count the nonzeros. The gap between those two counts is the whole problem. Chapter 18 explains why an approximate solve can be enough, and Chapter 33 explains what a sparsifier is.

Exercises

Exercise 3.18. Solve by elimination, showing the augmented matrix at each step:

$$2x + y - z = 0, \quad x - y + 2z = 9, \quad 3x + 2y + z = 7.$$

Verify your answer in all three original equations.

Exercise 3.19. For each system, decide without fully solving it whether there are no solutions, one, or infinitely many, and say how you know.

(a) $x + y + z = 1, \quad 2x + 2y + 2z = 3.$

(b) $x + y = 1, \quad x - y = 3, \quad 2x = 4.$

(c) $x + y + z = 0, \quad x - z = 0.$

Exercise 3.20. Find the complete solution set of $x_1 + 2x_2 + 3x_3 + 4x_4 = 5$, written in the particular-plus-homogeneous form of Theorem 3.14. How many free parameters are there, and why does Theorem 3.10 predict that number?

Exercise 3.21. For which values of λ does the system

$$x + 2y = 1, \quad 3x + \lambda y = 2$$

have a unique solution, and for which does it have no solution? One of the three outcomes of Theorem 3.2 never occurs here; identify it and explain why. Draw the cases you do find.

Exercise 3.22. Let A be 3×5 . Explain why $Ax = 0$ must have a nonzero solution, and give the smallest possible number of free parameters. Then construct an A that attains it.

Exercise 3.23. Show that if u and v both solve $Ax = b$, then $\lambda u + (1 - \lambda)v$ solves it for every $\lambda \in \mathbb{R}$, but $u + v$ solves it only when $b = 0$. Interpret both statements using Figure 3.4.

Exercise 3.24. Bring the matrix

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 8 \\ 1 & 2 & 2 & 5 \end{pmatrix}$$

to reduced row echelon form. Identify the pivot columns and the free columns, and state the rank.

Exercise 3.25. The same $n \times n$ matrix A must be used to solve $Ax = b$ for k different right-hand sides. Strategy one runs elimination from scratch each time, at about $\frac{2}{3}n^3$ operations per solve. Strategy two computes A^{-1} once, at about $2n^3$ operations, and then multiplies, at about $2n^2$ operations per solve. For large n , at roughly what k does strategy

two overtake strategy one? Library routines do neither: they factor A once at about $\frac{2}{3}n^3$ and then solve each system at about $2n^2$. Explain why that beats both.

Exercise 3.26. Theorem 3.9 asserts uniqueness for the reduced form only. For

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{pmatrix},$$

find two different sequences of elementary row operations reaching two different matrices in row echelon form in the sense of Definition 3.6, and then check that reducing each one further gives the same result. Where in the definition is the freedom that allows this?

References and further reading

The attribution of elimination to the *Nine Chapters* and the account of how Gauss's name became attached to it follow Grcar [272], which is the careful modern survey of the question; Chemla and Guo [142] give a scholarly edition of the *fangcheng* chapter itself. Our development of echelon form, the trichotomy and the particular-plus-homogeneous theorem is written from scratch for this book, but readers who want a slower, example-first treatment of the same ground will find Hefferon [307] excellent, and Austin [35] keeps the row and column pictures side by side throughout, which is the habit this chapter is trying to build. The uniqueness of reduced row echelon form is proved in both. Banica [51], our baseline for the later parts, treats systems briskly and moves quickly to the matrix-theoretic view, which is where Chapter 4 heads. For everything about pivoting, stability and what elimination actually does in floating-point arithmetic, the standard reference is Trefethen and Bau [607], whose lectures on Gaussian elimination and stability are the natural sequel to the final section here.